Marginal Persistence and Dynamic Copula Dependence in Sovereign Rating Migration Counts: A Discrete Interval-Likelihood MAGMAR Analysis

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Abstract. This paper develops an observed-data likelihood for applying moving-aggregate modified autoregressive (MAGMAR) copula time-series models to discrete sovereign rating-migration counts with time-varying exposure. An annual count identifies a probability-integral-transform interval rather than a unique latent point, so the likelihood integrates the latent process over the complete sequence of count intervals. A guided sequential Monte Carlo implementation incorporates the parameter-dependent stationary marginal adjustment of MAGMAR directly inside this integration. In sovereign ratings from Fitch, Moody's, and S&P over 1963–2025, the migration counts are strongly overdispersed relative to a binomial margin and, under an exposure-conditioned static beta-binomial margin, MAGMAR improves substantially on MAG. The interpretation changes once the marginal mean is allowed to depend on the lagged migration rate: residual serial correlation in the count margin disappears, a reference dynamic-margin fit gives only a small MAGMAR improvement, and information criteria favor MAG. Numerical profiles, higher-precision refits, and parameter-recovery experiments show that marginal persistence and copula persistence are only weakly separated in this short discrete series. The results therefore support interval integration as the correct observation layer while showing that conclusions about latent dynamic dependence must be conditioned on the specification of the count margin.

Keywords: copula time series, sovereign credit risk, rating migrations, discrete time series, beta-binomial, sequential Monte Carlo, MAGMAR

1 Introduction

Sovereign rating migrations affect bond portfolios, bank and insurance balance sheets, sovereign-backed lending, eligibility rules, and migration-based measures of credit risk. Most quantitative work models the probability or intensity of a transition between rating states. The present paper asks a different time-series question: after accounting for the number of country–agency relationships exposed to a possible migration, does aggregate annual migration activity display persistent dynamic dependence?

Two features make this question nonstandard. First, the annual number of eligible country–agency pairs changes sharply through time. A count of 20 migrations has a different interpretation when 60 pairs are at risk than when 300 are at risk. Second, migration counts are discrete. If the

conditional count distribution in year t has CDF F_t, observing $A_t = a_t$ implies only that a latent uniform variable lies in

$$\big(F_t(a_t^-), F_t(a_t)\big],$$

rather than revealing a unique probability-integral-transform value. A continuous copula likelihood evaluated at one jittered point inside this interval is therefore not the observed-data likelihood of the count sequence.

The methodological contribution of the paper is to construct an explicit discrete observation layer for the MAG and MAGMAR copula processes of Pappert (2026). The latent recursion is retained, but the observation map is formulated through exposure-conditioned count intervals. For each candidate parameter vector, the stationary marginal CDF of the raw MAGMAR state is estimated, the observed count intervals are mapped back to raw-state intervals, and a guided particle filter integrates the conditional latent density over those intervals. This produces a simulated approximation to the observed-count rectangle likelihood.

The empirical contribution is to separate two forms of persistence that can look similar in a short discrete series. Under a static beta-binomial margin, Gaussian MAGMAR is much better than Gaussian MAG: the publication fit gives log likelihoods -170.69 and -178.88, respectively. The fitted MAGMAR dependence is characterized by a strong positive autoregressive copula component and a negative moving-aggregate component. Yet the static count margin leaves substantial residual serial correlation. Once the beta-binomial mean is allowed to depend on the previous year's migration rate, the residual Ljung–Box statistic falls from 102.43 to 10.04 at lag 10, with nominal $p = 0.437$ for the dynamic margin. Under this richer observation model, a reference joint fit yields only 0.77 log-likelihood units of improvement from MAGMAR over MAG, while AIC and BIC favor MAG.

The dynamic-margin results also reveal a finite-sample identification issue that is substantively important rather than merely computational. Independent profile likelihoods rise toward large positive values of the MAGMAR autoregressive parameter, and higher-precision refits can move from a moderate-ϕ solution to a high-ϕ solution. Parameter-recovery simulations reproduce the margin coefficients well but recover ϕ and θ only weakly. Low-exposure sensitivities likewise do not produce a stable MAGMAR advantage. The empirical message is therefore not that one dependence specification dominates under every margin. It is that discrete interval integration changes the inferential object, and that marginal dynamics must be modeled carefully before persistent copula dependence is interpreted structurally.

The paper makes four contributions:

1. it formulates an exposure-conditioned discrete observation model for MAG/MAGMAR in which counts identify latent probability intervals;

2. it develops a guided sequential Monte Carlo approximation to the resulting observed-count likelihood, including the parameter-dependent stationary marginal adjustment inside the likelihood calculation;

3. it compares static and dynamic beta-binomial observation layers and shows how marginal

persistence changes the apparent evidence for latent dependence;

4. it provides numerical diagnostics for Monte Carlo error, profile geometry, probability-grid resolution, finite-sample parameter recovery, influential years, and direction-specific behavior.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the sovereign ratings, exposure, and beta-binomial observation models. Section 4 develops the discrete MAG/MAGMAR likelihood. Section 5 describes simulation-based estimation and the validation design. Section 6 reports the empirical results, Section 7 discusses interpretation, and Section 8 concludes.

2 Related literature and methodological background

This paper connects three literatures: credit-rating transitions, copula models for discrete time series, and simulation-based likelihood evaluation.

Rating-transition models commonly use discrete- or continuous-time Markov structures. Jarrow et al. (1997) develop a Markov framework for credit-risk spreads, while Lando and Skødeberg (2002) study rating transitions and rating drift using continuous observations. Bangia et al. (2002) show that migration behavior changes with the business cycle, and Figlewski et al. (2012) analyze macroeconomic effects on default and rating-transition dynamics. Those models are designed around transition probabilities, intensities, or transition matrices. Here the empirical object is the aggregate number of annual sovereign rating migrations, modeled as a univariate discrete time series conditional on the number of rating relationships exposed to a possible migration.

Copula models separate marginal behavior from dependence through Sklar's theorem (Sklar, 1959); see Joe (2014) for a comprehensive treatment. This separation is attractive when the marginal distribution changes mechanically with exposure. Discrete margins, however, require care. For integer-valued observations the copula is observed only on the range of the marginal CDF, so the latent uniform is interval-censored rather than observed exactly. Genest and Nešlehová (2007) discuss the resulting identification issues for count data, while Trivedi and Zimmer (2017) emphasize that covariate variation, and hence variation in observable CDF grids, can help distinguish parametric dependence structures. Weiss (2018) provides a broader treatment of discrete-valued time series.

The latent dynamic model is the MAGMAR copula process of Pappert (2026). MAGMAR augments an autoregressive copula recursion with a moving-aggregate component driven by current and lagged innovations. A key feature for the present application is that the stationary marginal distribution of the raw MAGMAR process need not be uniform. The stationary marginal transformation that restores uniformity depends on the candidate dependence parameters. In a discrete observation model this transformation therefore changes the raw-state intervals associated with each observed count and must be recomputed inside the likelihood.

The resulting observed-data likelihood is a high-dimensional rectangle probability for a nonlinear latent process and is not available in closed form. We evaluate it sequentially using particle

importance sampling and systematic resampling, following the general SMC logic developed for nonlinear and non-Gaussian state-space models; see Del Moral (2004) and Doucet and Johansen (2011). Common random numbers are used within an optimization run so that the simulated objective is reproducible and substantially smoother than an objective based on independent simulation at every parameter evaluation.

3 Data and the discrete observation layer

3.1 Sovereign ratings and annual migration counts

The empirical application uses sovereign credit ratings from Fitch, Moody's, and S&P obtained from TheGlobalEconomy. Ratings are standardized to an ordered numeric scale on which larger values denote stronger credit quality. The raw information is converted to a country–agency–year panel. For each country–agency pair, the annual rating is compared with its preceding eligible annual rating. A downgrade occurs when the current rating is lower, an upgrade when it is higher, and a migration when either event occurs.

The modeled observation is the number of eligible country–agency pairs whose annual rating differs from the preceding annual rating, rather than the number of every within-year rating announcement. The construction starts from 9,748 raw rating rows; 187 NR/WR/SD/RD rows and 29 rows with unmapped rating codes are excluded. The resulting annual panel contains 10,290 country–agency–year rows and 459 country–agency pairs. There are 9,831 eligible annual comparisons and 2,109 migrations, comprising 952 downgrades and 1,157 upgrades.

Let E_t denote the number of country–agency pairs eligible for a migration comparison in year t. Let N_t^- and N_t^+ denote the numbers of downgrades and upgrades, respectively, and define

$$A_t = N_t^- + N_t^+. \tag{1}$$

Pairs with fewer than three distinct annual observations are removed before aggregation, so the minimum-history rule is applied at the country–agency level. The aggregate series contains 63 annual observations from 1963 through 2025. The migration count ranges from 0 to 101 and exposure from 1 to 390.

3.2 Static and dynamic beta-binomial margins

Because A_t counts migrations among E_t eligible pairs,

$$A_t \in \{0, 1, \ldots, E_t\}.$$

For a generic exposure-conditioned discrete margin write

$$F_t(a;\gamma) = \mathbb{P}_\gamma(A_t \le a \mid E_t, \mathcal{F}_{t-1}), \qquad m_t(a;\gamma) = \mathbb{P}_\gamma(A_t = a \mid E_t, \mathcal{F}_{t-1}).$$

Table 1: Construction audit and aggregate sovereign migration sample.

Quantity	Value
Raw rating rows	9,748
Excluded NR/WR/SD/RD rows	187
Rows with unmapped rating codes	29
Country–agency–year rows in annual panel	10,290
Country–agency pairs	459
Eligible annual comparison rows	9,831
Total migrations	2,109
Downgrades	952
Upgrades	1,157
Aggregate sample	1963–2025
Number of annual observations	63
Range of migration count A_t	0–101
Range of exposure E_t	1–390

The observation layer is beta-binomial. Conditional on a migration probability $p_t \in (0,1)$ and precision parameter $\kappa > 0$,

$$A_t \mid E_t, \mathcal{F}_{t-1} \sim \text{BetaBinomial}(E_t, p_t, \kappa), \tag{2}$$

with beta shape parameters $\alpha_t = p_t\kappa$ and $\beta_t = (1-p_t)\kappa$. Its probability mass is

$$m_t(a; p_t, \kappa) = \binom{E_t}{a} \frac{B(a + p_t\kappa, E_t - a + (1-p_t)\kappa)}{B(p_t\kappa, (1-p_t)\kappa)}, \tag{3}$$

and

$$\mathbb{E}(A_t \mid E_t, \mathcal{F}_{t-1}) = E_t p_t, \tag{4}$$

$$\text{Var}(A_t \mid E_t, \mathcal{F}_{t-1}) = E_t p_t (1-p_t) \frac{E_t + \kappa}{1+\kappa}. \tag{5}$$

Two specifications are used. The *static* beta-binomial margin sets

$$\text{logit}(p_t) = \beta_0. \tag{6}$$

The *dynamic* margin uses the lagged observed migration rate,

$$\text{logit}(p_t) = \beta_0 + \beta_1 R_{t-1}, \qquad R_{t-1} = \frac{A_{t-1}}{E_{t-1}}. \tag{7}$$

The dynamic analysis therefore uses 62 observations from 1964–2025. Setting $\beta_1 = 0$ recovers the static mean specification on the same sample.

For every admissible count define the CDF interval

$$\ell_t(a;\gamma) := F_t(a^-;\gamma), \tag{8}$$

$$r_t(a;\gamma) := F_t(a;\gamma), \tag{9}$$

$$I_t(a;\gamma) := (\ell_t(a;\gamma), r_t(a;\gamma)]. \tag{10}$$

Its width is exactly the probability mass,

$$|I_t(a;\gamma)| = m_t(a;\gamma). \tag{11}$$

Thus an observed count identifies a region of latent probability space rather than a single latent point.

Proposition 3.1 (Exposure-conditional distributional transform)**.** *Suppose $A_t \mid E_t, \mathcal{F}_{t-1}$ has CDF $F_t(\cdot;\gamma_0)$ and let $V_t \sim \mathrm{Unif}(0,1)$ be independent of the count process. Then*

$$U_t^{(V)} = F_t(A_t^-;\gamma_0) + V_t\, m_t(A_t;\gamma_0) \tag{12}$$

is conditionally uniform on $(0,1)$ and satisfies $A_t = F_t^{-1}(U_t^{(V)};\gamma_0)$ almost surely.

Proof. Conditional on the observed conditioning variables, the intervals $I_t(a;\gamma_0)$ partition $(0,1]$ and have lengths equal to the corresponding conditional count probabilities. Conditional on $A_t = a$, expression (12) is uniform on $I_t(a;\gamma_0)$. Mixing these conditional uniform distributions with weights $m_t(a;\gamma_0)$ therefore produces a uniform distribution on $(0,1)$. The inverse-CDF statement follows from $U_t^{(V)} \in I_t(A_t;\gamma_0)$ almost surely. □

The distributional transform is useful for diagnostics, but the observed-data likelihood below integrates over the full interval rather than conditioning on one realization of V_t.

4 Discrete MAG and MAGMAR interval likelihood

4.1 Gaussian MAG(1) and MAGMAR(1,1)

Let $\{W_t\}_{t\in\mathbb{Z}}$ be iid $\mathrm{Unif}(0,1)$. For a bivariate copula C_ρ define

$$h_\rho(u,v) = \frac{\partial}{\partial u} C_\rho(u,v),$$

and let $h_\rho^{-1}(\cdot,v)$ denote inversion in the first argument. The empirical implementation uses Gaussian copulas for the autoregressive and moving-aggregate components.

The Gaussian MAG(1) process is

$$U_t = h_\theta^{-1}(W_t, W_{t-1}), \tag{13}$$

where $\theta \in (-1, 1)$. The Gaussian MAGMAR(1,1) process is

$$U_t = h_\phi^{-1}\Big(h_\theta^{-1}(W_t, W_{t-1}), U_{t-1}\Big)\,, \tag{14}$$

where $\phi \in (-1, 1)$ is the autoregressive copula parameter and $\theta \in (-1, 1)$ is the moving-aggregate parameter. Setting $\phi = 0$ reduces MAGMAR(1,1) to MAG(1).

The conditional density contribution of the raw Gaussian MAGMAR state is

$$f_{\phi,\theta}(u_t \mid u_{t-1}, w_{t-1}) = c_\phi(u_t, u_{t-1})\, c_\theta(h_\phi(u_t, u_{t-1}), w_{t-1})\,, \tag{15}$$

where c_ρ is the Gaussian copula density. The innovation state is updated by

$$w_t = h_\theta(h_\phi(u_t, u_{t-1}), w_{t-1})\,.$$

For MAG(1), the density contribution reduces to $c_\theta(u_t, w_{t-1})$ and $w_t = h_\theta(u_t, w_{t-1})$.

4.2 Stationary marginal adjustment

A raw MAGMAR process generated by (14) need not have a uniform stationary marginal (Pappert, 2026). Let

$$\Psi_\eta(u) = \mathbb{P}_\eta(U_t \le u), \qquad \eta = (\phi, \theta)', \tag{16}$$

denote the stationary marginal CDF of the raw latent process. When Ψ_η is continuous,

$$\widetilde{U}_t = \Psi_\eta(U_t) \tag{17}$$

is marginally uniform. The observed discrete process is defined by

$$A_t = F_t^{-1}(\widetilde{U}_t; \gamma). \tag{18}$$

Thus observing $A_t = a_t$ is equivalent to $\widetilde{U}_t \in I_t(a_t; \gamma)$ and, in the raw MAGMAR state,

$$U_t \in J_t(a_t; \eta, \gamma) := \Big(\Psi_\eta^{-1}\{\ell_t(a_t; \gamma)\}, \Psi_\eta^{-1}\{r_t(a_t; \gamma)\}\Big]\,. \tag{19}$$

The inverse stationary transformation is therefore part of every likelihood evaluation.

4.3 Observed-count rectangle likelihood

Let $\xi = (\eta', \gamma')'$ collect dependence and margin parameters and define

$$\mathcal{I}_T(a_{1:T}; \gamma) = \prod_{t=1}^{T} I_t(a_t; \gamma).$$

If $\widetilde{g}_{\eta,T}$ is the joint density of $(\widetilde{U}_1, \ldots, \widetilde{U}_T)$, the observed-data likelihood is

$$L_T^D(\xi) = \int_{\mathcal{I}_T(a_{1:T};\gamma)} \widetilde{g}_{\eta,T}(u_{1:T})\, du_{1:T}. \tag{20}$$

Equivalently, this is the probability under the raw MAGMAR process that all U_t fall inside the parameter-dependent intervals in (19).

Theorem 4.1 (Randomized integral representation)**.** *Suppose $\widetilde{g}_{\eta,T}$ exists and is integrable. For iid $V_t \sim \mathrm{Unif}(0,1)$ define*

$$u_t(V_t;\gamma) = \ell_t(a_t;\gamma) + V_t m_t(a_t;\gamma).$$

Then

$$L_T^D(\eta,\gamma) = \left\{\prod_{t=1}^{T} m_t(a_t;\gamma)\right\} \mathbb{E}_V\left[\widetilde{g}_{\eta,T}\{u_1(V_1;\gamma), \ldots, u_T(V_T;\gamma)\}\right]. \tag{21}$$

Proof. Apply the coordinatewise change of variables $u_t = \ell_t(a_t;\gamma) + v_t m_t(a_t;\gamma)$ to the rectangle integral (20). The Jacobian determinant is $\prod_{t=1}^{T} m_t(a_t;\gamma)$. □

The order of averaging and taking logarithms matters:

$$\log \mathbb{E}_V\{\widetilde{g}_{\eta,T}(u(V;\gamma))\} \neq \mathbb{E}_V\{\log \widetilde{g}_{\eta,T}(u(V;\gamma))\}.$$

Hence a continuous likelihood evaluated at one randomized transform is a different criterion from the observed-count likelihood in (20).

4.4 Identification and changing exposure

For a discrete margin define the probability grid

$$\mathcal{G}_t(\gamma) = \{0, F_t(0;\gamma), \ldots, F_t(E_t;\gamma) = 1\}$$

and its mesh

$$\Delta_t(\gamma) = \max_{0 \le a \le E_t} m_t(a;\gamma). \tag{22}$$

The observed discrete process reveals latent probabilities only on products of these grids.

Assumption 4.2 (Refining grids and block identification)**.** (i) the marginal parameter γ_0 is identified from the exposure-conditioned one-dimensional distributions;

(ii) for each fixed block length $d \ge 2$, there exists a sequence of consecutive blocks B_k such that

$$\max_{t \in B_k} \Delta_t(\gamma_0) \longrightarrow 0;$$

(iii) the adjusted latent d-dimensional CDF is continuous;

(iv) within the specified parametric MAGMAR family, equality of all finite block distributions implies equality of the dependence parameter.

Theorem 4.3 (Parametric identification under refining exposure grids)**.** *Under Assumption 4.2, if two parameter vectors* $\xi_1 = (\eta_1, \gamma_1)$ *and* $\xi_2 = (\eta_2, \gamma_2)$ *generate the same finite-dimensional distributions for the observed count process conditional on the refining exposure blocks, then* $\xi_1 = \xi_2$.

Proof. Equality of the one-dimensional count distributions identifies $\gamma_1 = \gamma_2 = \gamma_0$. For a fixed block length d, equality of observed block distributions implies equality of adjusted latent block CDFs at every point of the corresponding product probability grid. By the refining-mesh condition, the union of these grids is dense in $[0,1]^d$. Continuity extends equality from the grids to all of $[0,1]^d$, and block identification of the parametric MAGMAR family then yields $\eta_1 = \eta_2$. □

The theorem is a population statement. In the observed sample, exposure equals one for many early years and later rises above 300, so grid resolution changes markedly through time. Section 6.5 evaluates that finite-sample resolution directly through the empirical mesh, profile objectives, low-exposure subsamples, and parameter recovery.

5 Simulated likelihood and validation design

5.1 Guided sequential Monte Carlo likelihood

For the static margin, the parameter vector is

$$\xi_{\text{MAGMAR}} = (\beta_0, \log\kappa, \phi, \theta)', \qquad \xi_{\text{MAG}} = (\beta_0, \log\kappa, \theta)'.$$

For the dynamic margin, β_1 is added. For each candidate parameter vector, the likelihood calculation proceeds as follows.

1. **Count intervals.** Construct the beta-binomial intervals $I_t(A_t; \gamma)$ under the candidate static or dynamic margin.
2. **Stationary marginal map.** For MAGMAR, simulate a long stationary raw path at the candidate (ϕ, θ) and construct numerical maps $\widehat{\Psi}_\eta$ and $\widehat{\Psi}_\eta^{-1}$.
3. **Raw-state intervals.** Map the count intervals back through $\widehat{\Psi}_\eta^{-1}$. For MAG, whose raw construction is uniform in the implementation, the count intervals are used directly.
4. **Initialization.** The first count contributes its fitted marginal probability. A stationary simulation pool is conditioned on the first raw-state interval to initialize particles for (U_1, W_1).
5. **Sequential integration.** For each subsequent year, particles propose U_t uniformly over the current raw-state interval. Importance weights equal the interval width times the conditional density in (15). Weights are normalized and systematic resampling propagates particles and innovation states.

6. **Optimization.** The sum of sequential log normalizing constants is the simulated log likelihood. Common random numbers are held fixed across parameter evaluations within a run. BFGS optimization is used with multiple starts.

The saved publication fits reported in Section 6 use 6,000 particles and an 80,000-state stationary pool. The lighter validation exercises use 2,000 particles and a 20,000-state pool unless stated otherwise, with stationary burn-in 3,000, BFGS maximum iteration count 250, and relative tolerance 2×10^{-5}. Direction-specific and influential-year refits use 1,500 particles and 16,000-state pools.

5.2 Implementation and Monte Carlo checks

The implementation contains four direct unit checks. The Gaussian h-function and its inverse agree to machine precision over several dependence parameters. Simulated MAG states are marginally uniform, and the stationary-adjusted MAGMAR states are uniform after applying the estimated stationary map. Under MAG independence ($\theta = 0$), the interval likelihood agrees with the product of marginal interval masses to numerical precision. A $T = 2$ brute-force integration check agrees with the SMC likelihood within 1.61×10^{-3}.

Monte Carlo stability is assessed in two ways. First, the fitted dynamic MAG and MAGMAR parameters are re-evaluated on 12 independent streams for each combination of particle counts 2,000, 4,000, 8,000, and 12,000 and stationary-pool sizes 20,000, 50,000, and 100,000. Second, four independent full refits at the default validation configuration quantify simulation-induced variation in both likelihood differences and parameter estimates.

5.3 Profiles, bootstrap, and finite-sample recovery

The dynamic-margin specification is used for the main finite-sample validation because it directly addresses the strongest alternative explanation for apparent copula persistence. Independent-stream nuisance profiles are computed for ϕ and θ. A parametric bootstrap simulates count series under fitted dynamic MAG on the actual exposure path and refits MAG and MAGMAR. Parameter recovery simulates from the reference dynamic MAGMAR fit on the same exposure path and re-estimates all parameters.

The bootstrap is used as a numerical calibration exercise rather than as a source of high-precision tail probabilities. With only 49 replications, its Monte Carlo resolution is coarse. In addition, the stored run contains one obvious null-fit optimization failure, producing an LR statistic above 1,500, and several negative computed LR values caused by the alternative optimizer not fully recovering the nested null solution. These features are reported transparently in Section 6.3; they do not support a conventional Wilks-style significance claim.

Table 2: Beta-binomial margin diagnostics. The static fit uses 1963–2025; the dynamic fit begins in 1964 because it conditions on the lagged migration rate.

Margin	n	$\log L$	AIC	$\widehat{\kappa}$	Residual SD	LB(10) p
Static beta-binomial	63	-189.792	383.584	44.703	1.019	$< 10^{-16}$
Dynamic beta-binomial	62	-174.015	354.029	104.373	1.125	0.437

Table 3: Static beta-binomial observed-count interval-SML fits.

Model	k	$\log L_{\mathrm{SML}}$	AIC	BIC	$\widehat{\kappa}$	$\widehat{\phi}$	$\widehat{\theta}$
MAG(1)	3	-178.876	363.752	370.181	63.238	—	0.404
MAGMAR(1,1)	4	-170.688	349.376	357.949	89.026	0.879	-0.530

6 Empirical results

6.1 Overdispersion and marginal persistence

The static beta-binomial margin has fitted precision $\widehat{\kappa} = 44.70$ and substantially improves on the exposure-only binomial margin used in preliminary analysis. More importantly for the dependence analysis, its Pearson residuals retain strong serial structure: the Ljung–Box statistic at lag 10 is 102.43, with $p \approx 1.8 \times 10^{-17}$. The dynamic beta-binomial margin changes this diagnostic sharply. With

$$\widehat{\beta}_0 = -2.205, \qquad \widehat{\beta}_1 = 4.105, \qquad \widehat{\kappa} = 104.37,$$

the Ljung–Box statistic falls to 10.04 and the nominal p-value rises to 0.437. The dynamic count margin therefore absorbs essentially all residual linear serial correlation detected by this diagnostic.

Because the two rows use different starting years, the likelihood criteria in Table 2 are descriptive rather than a formal nested test. The serial-diagnostic change is the central point: a static observation mean leaves a large predictable component in the counts, whereas a one-lag migration-rate term removes it.

6.2 Static-margin interval likelihood strongly favors MAGMAR

Table 3 reports the saved publication fits under the static beta-binomial margin. Gaussian MAGMAR(1,1) has simulated observed-count log likelihood -170.688, compared with -178.876 for Gaussian MAG(1). The descriptive nested statistic is

$$2(-170.688 + 178.876) = 16.376.$$

AIC falls from 363.75 to 349.38 and BIC from 370.18 to 357.95. The MAGMAR fit has a large positive autoregressive parameter, $\widehat{\phi} = 0.879$, and a negative moving-aggregate parameter, $\widehat{\theta} = -0.530$.

The static-margin result is clear as a statement about that observation model. It is not sufficient by itself for a structural interpretation of latent persistence, because the static beta-binomial

Table 4: Reference dynamic beta-binomial interval-SML fits on 1964–2025.

Model	k	$\log L_{\mathrm{SML}}$	AIC	BIC	$\widehat{\beta}_1$	$\widehat{\kappa}$	$\widehat{\phi}$	$\widehat{\theta}$
MAG(1)	4	-173.888	355.776	364.284	3.645	104.194	—	0.079
MAGMAR(1,1)	5	-173.119	356.238	366.874	4.308	92.008	0.684	-0.502

residuals remain serially correlated. This is why the dynamic margin is treated as the central specification sensitivity rather than as an optional appendix check.

6.3 Dynamic margin: a much weaker and less sharply identified MAGMAR increment

On the common 1964–2025 sample, the reference dynamic-margin MAG fit has log likelihood -173.888. The reference MAGMAR fit improves this to -173.119, a gain of only 0.769 log-likelihood units and an LR statistic of 1.537. With one additional parameter, MAGMAR has slightly higher AIC and BIC than MAG. The corresponding reference dependence estimates are

$$\widehat{\phi} = 0.684, \qquad \widehat{\theta} = -0.502.$$

The reference pair should not be read as establishing a unique global maximum for the dynamic MAGMAR objective. Independent nuisance profiles reveal a higher-ϕ region. Averaging over two independent likelihood streams, the profile log likelihood rises from -172.94 at $\phi = 0.684$ to -171.35 at $\phi = 0.90$. Higher-precision full refits display the same behavior: the MAG fit remains near -173.88, whereas the MAGMAR solution can move toward $\phi \approx 0.93$ and $\theta \approx -0.63$. This is evidence of a broad or multimodal dependence surface, not a basis for selecting one numerical mode as a precise estimator.

The 49-replication parametric bootstrap around the reference dynamic MAG fit is also non-decisive. The observed LR is 1.537. Retaining all stored replications gives a plus-one upper-tail probability of 0.12. One replication is an obvious numerical failure: the null MAG refit lands at log likelihood -943.7 while the MAGMAR refit is near -165.6, producing an LR above 1,500 despite a nominal convergence flag. Treating this run as invalid leaves 48 replications and a plus-one exceedance rate of approximately 0.10. Several negative computed LRs further indicate optimizer noise in the nested comparison. The bootstrap therefore supports the qualitative conclusion that the reference dynamic MAGMAR increment is not strongly significant, but it is not used for precise tail inference.

6.4 Monte Carlo stability and profile geometry

At the fixed reference parameters, paired independent-stream evaluations are much more stable than the full refits. Across 12 particle/pool configurations, the mean MAGMAR–MAG log-likelihood difference ranges from 0.395 to 0.690. The share of replications with a positive difference ranges

Table 5: Selected paired Monte Carlo diagnostics at the reference dynamic-margin parameters. Each row uses 12 independent streams.

Particles	Pool	Mean $\Delta\ell$	SD $\Delta\ell$	MCSE(mean)	Share $\Delta\ell > 0$
2,000	20,000	0.690	0.493	0.142	0.917
4,000	100,000	0.582	0.115	0.033	1.000
8,000	100,000	0.592	0.159	0.046	1.000
12,000	100,000	0.497	0.126	0.036	1.000

from 0.833 to 1.000, and the mean MAGMAR median ESS is essentially equal to the particle count in every configuration. Thus the particle filter itself is not collapsing around the reference solution.

Four independent full refits at the default validation configuration give mean $\Delta\ell = 1.316$ with MCSE 0.235. The mean dependence estimates are $\widehat{\phi} = 0.691$ and $\widehat{\theta} = -0.512$, with MCSEs 0.012 and 0.023, respectively. The margin estimates are similarly stable. The contrast between stable local refits and a profile that improves at high ϕ is characteristic of a multimodal or ridge-like objective: simulation noise is manageable locally, while global separation of the dependence components is weaker.

6.5 Finite-sample grid resolution and parameter recovery

The empirical probability-grid mesh confirms the role of changing exposure in Theorem 4.3. Under the reference dynamic MAGMAR margin, the largest cell width has median 0.077, minimum 0.021, and maximum 0.904. The correlation between exposure and the maximum grid-cell width is -0.716. The very coarse early years are therefore fundamentally less informative about latent probabilities than the later high-exposure years.

Table 6: Finite-sample parameter recovery on the actual exposure path, 20 replications.

Parameter	True value	Bias	RMSE	2.5%–97.5% of estimates
ϕ	0.684	-0.329	0.470	$[-0.133,\ 0.793]$
θ	-0.502	0.324	0.475	$[-0.763,\ 0.341]$
β_0	-2.241	0.006	—	—
β_1	4.308	-0.004	—	—
κ	92.008	-1.799	—	—

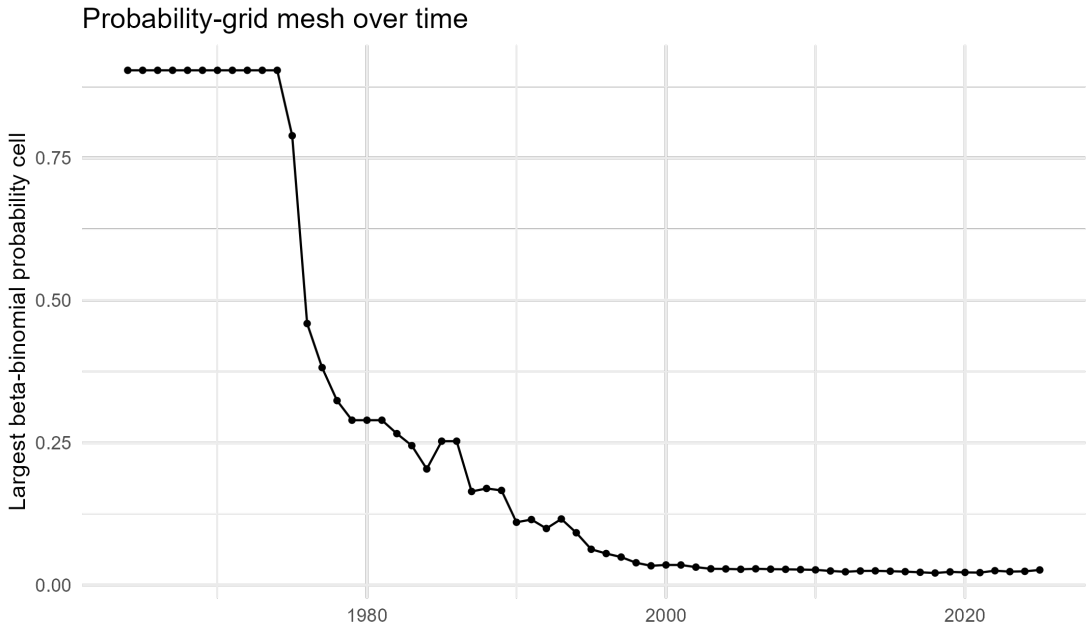


Figure 1: Maximum beta-binomial probability-grid cell width by year under the reference dynamic-margin fit. Larger exposure produces a finer observable probability grid.

Parameter-recovery experiments make the finite-sample implication explicit. Twenty series are simulated from the reference dynamic MAGMAR fit on the actual exposure path and then re-estimated. All runs return convergence flags and none are classified as boundary solutions. The margin parameters recover well: biases are 0.006 for β_0, -0.004 for β_1, and -1.80 for κ. Dependence recovery is much weaker. The bias and RMSE are -0.329 and 0.470 for ϕ, and 0.324 and 0.475 for θ.

The pattern is informative: the exposure-conditioned mean is recoverable at this sample size, while the decomposition of serial dependence into ϕ and θ is not sharply resolved. This is consistent with the profile geometry and with the change in model ranking after adding the lagged count rate to the margin.

Low-exposure subsamples reinforce the same conclusion. Starting the dynamic analysis when exposure first reaches 25, 50, or 100 produces descriptive MAGMAR–MAG LR statistics of approximately -0.01, 2.84, and -4.35, respectively. These signs are not interpreted literally as nested LR

statistics because the simulated optimization can return a lower alternative objective. The relevant result is that there is no stable MAGMAR advantage across exposure thresholds.

6.6 Influential years and directional decomposition

The largest dynamic beta-binomial Pearson residuals occur in 1975 and 1974, when exposure is extremely small. Masking either year, or both together, leaves the reference MAGMAR dependence parameters close to the full-sample local solution. Masking both gives $\widehat{\phi} = 0.684$ and $\widehat{\theta} = -0.502$ and a descriptive MAGMAR–MAG LR of 1.34. The local dependence pattern is therefore not generated by a single exceptional early observation.

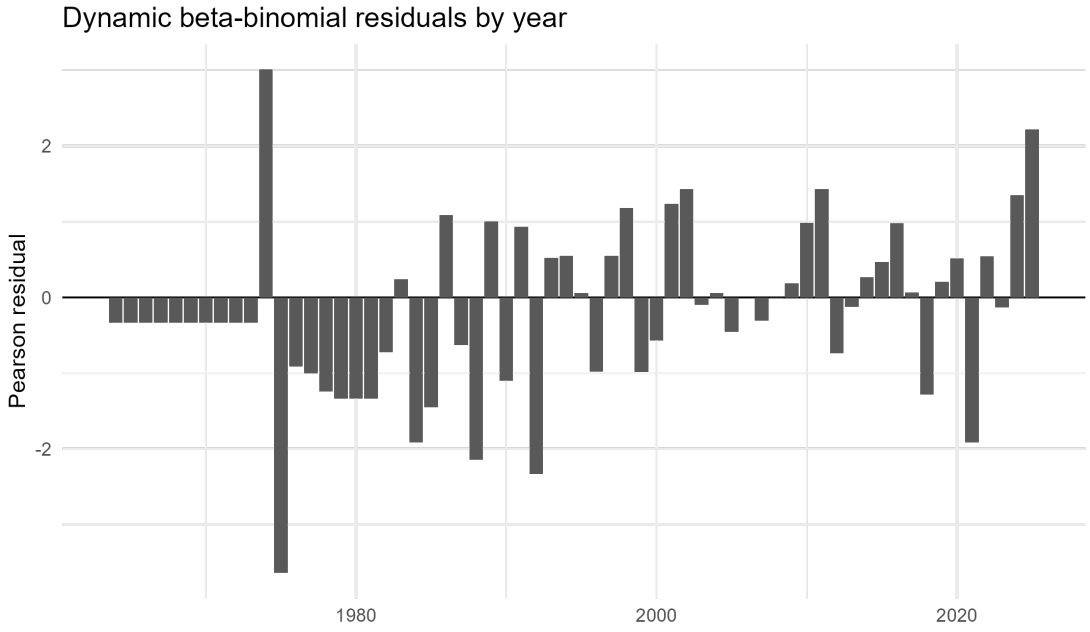


Figure 2: Pearson residuals from the dynamic beta-binomial count margin. The two largest absolute residuals occur in 1975 and 1974, when exposure is minimal.

Direction-specific interval fits are exploratory because splitting total migration activity into upgrades and downgrades reduces the effective information further. Under static beta-binomial margins, MAG outperforms MAGMAR for downgrades, while MAGMAR improves the upgrade log likelihood by 1.64 units. The separate fits also push the MAG moving-aggregate parameter above 0.88 and produce very low minimum ESS in some years. These results suggest asymmetry between upgrade and downgrade activity, but they are not used for the primary structural comparison.

7 Discussion

The central methodological result is that the discrete observation layer matters. A sovereign migration count maps to an interval of latent probability values, and MAGMAR's stationary

marginal adjustment changes that interval as the dependence parameters change. Integrating over the interval is therefore part of the model, not a cosmetic correction to a continuous transformed-series likelihood.

The central empirical result is a separation problem between marginal and copula persistence. If the beta-binomial migration probability is static, the observed count likelihood strongly favors MAGMAR. Once the migration probability responds to the previous year's migration rate, the count margin itself explains nearly all of the residual serial correlation detected by a Ljung–Box diagnostic. The remaining increment from the reference MAGMAR solution is small, and the likelihood surface contains a high-ϕ region that is not well separated from the moderate-ϕ solution. In short annual data, richer marginal dynamics and richer latent dependence can substitute for one another.

This interaction explains why a single model-selection statistic is not the right summary of the application. Under the static margin, AIC and BIC strongly favor MAGMAR. Under the reference dynamic local solution, they favor MAG. Independent profiles and higher-precision refits then reveal that the dynamic MAGMAR objective can improve again at large ϕ. The numerical diagnostics are therefore part of the empirical result: they show that the data contain evidence of serial structure, but do not sharply allocate that structure between a lagged conditional mean and two copula dependence channels.

The parameter-recovery experiment provides the clearest finite-sample interpretation. The dynamic beta-binomial coefficients and precision parameter are recovered with small bias, whereas the dependence parameters have RMSEs near 0.47 and broad empirical ranges. This distinction is consistent with the observed probability-grid mesh. Later high-exposure years provide fine count intervals, but the series still contains only 62 dynamic observations, and the early part of the sample contributes very coarse probability information.

For credit-risk applications, the framework is most useful when the research question concerns aggregate migration activity rather than individual transition matrices. It complements transition-matrix and intensity models by providing a way to model persistent annual variation in the total amount of rating movement after conditioning on exposure. The analysis also gives a practical specification lesson: a latent dependence model should be compared against a dynamic count margin before persistence is assigned to the copula recursion.

8 Conclusion

This paper develops a discrete interval-likelihood framework for applying MAG and MAGMAR copula time-series models to sovereign rating-migration counts with changing exposure. The observed count identifies a CDF interval rather than a unique PIT value, and the MAGMAR stationary marginal transformation makes the corresponding raw latent interval parameter-dependent. A guided SMC algorithm integrates over these intervals directly.

In the sovereign application, a static beta-binomial margin produces strong evidence in favor of MAGMAR relative to MAG. That result changes once marginal persistence is modeled explicitly.

A beta-binomial mean driven by the lagged migration rate removes the residual serial correlation left by the static margin, and the reference dynamic MAGMAR increment becomes small. At the same time, independent profiles and higher-precision refits reveal a high-autoregressive-dependence region, while parameter recovery shows that ϕ and θ are weakly resolved in samples of this length.

The broader implication is twofold. First, discrete copula time-series models should be estimated on the probability intervals implied by the observations rather than on arbitrary randomized representatives of those intervals. Second, in short count series, evidence for latent serial dependence must be interpreted jointly with the dynamics allowed in the observation margin. For sovereign migration counts, the interval likelihood identifies persistent structure, but the available sample does not support a sharp decomposition of that persistence into marginal and MAGMAR components.

A Implementation unit tests

Table 7: Implementation checks used in the current validation run.

Test	Setting	Statistic	Pass
h-inverse identity	$\rho = -0.5$	4.44×10^{-16}	yes
h-inverse identity	$\rho = 0$	2.22×10^{-16}	yes
h-inverse identity	$\rho = 0.25$	2.50×10^{-16}	yes
h-inverse identity	$\rho = 0.6$	6.66×10^{-16}	yes
MAG marginal uniformity	$\theta = 0.35$	0.00334	yes
Adjusted MAGMAR uniformity	$\phi = 0.30, \theta = 0.20$	0.00420	yes
Independence interval identity	MAG $\theta = 0$	8.88×10^{-16}	yes
$T = 2$ brute force vs SMC	MAG $\theta = 0.35$	0.00161	yes

B Full paired Monte Carlo grid

Table 8: Paired independent-stream evaluation of the dynamic reference fits. Each configuration has 12 valid replications.

Particles	Pool	Mean $\Delta\ell$	SD	MCSE(mean)	Share positive
2,000	20,000	0.690	0.493	0.142	0.917
2,000	50,000	0.395	0.196	0.057	1.000
2,000	100,000	0.567	0.188	0.054	1.000
4,000	20,000	0.477	0.407	0.117	0.833
4,000	50,000	0.633	0.218	0.063	1.000
4,000	100,000	0.582	0.115	0.033	1.000
8,000	20,000	0.455	0.419	0.121	0.833
8,000	50,000	0.607	0.314	0.091	1.000
8,000	100,000	0.592	0.159	0.046	1.000
12,000	20,000	0.592	0.392	0.113	0.917
12,000	50,000	0.573	0.377	0.109	0.917
12,000	100,000	0.497	0.126	0.036	1.000

C Independent-stream profile summaries

Table 9: Dynamic MAGMAR nuisance profiles averaged over two independent likelihood streams. Relative likelihoods are normalized separately within each fixed parameter.

Fixed parameter	Fixed value	Mean profile $\log L$	Relative $\log L$
ϕ	0.000	-176.616	-5.261
ϕ	0.384	-174.042	-2.688
ϕ	0.513	-174.607	-3.253
ϕ	0.642	-173.455	-2.101
ϕ	0.684	-172.939	-1.585
ϕ	0.771	-172.629	-1.275
ϕ	0.900	-171.354	0.000
θ	-0.800	-173.642	-0.856
θ	-0.651	-178.855	-6.070
θ	-0.502	-172.785	0.000
θ	-0.501	-173.034	-0.249
θ	-0.352	-175.804	-3.019
θ	-0.202	-174.280	-1.495

The large between-stream standard deviation at $\theta = -0.651$ and the monotone improvement of the ϕ profile toward the top of the evaluated grid reinforce the interpretation of a broad and numerically challenging dependence surface.

D Low-exposure and influential-year sensitivity

Table 10: Dynamic-margin low-exposure sensitivity. LR values are descriptive because simulated optimization can violate the exact nesting inequality.

Exposure threshold	Start	n	ℓ_{MAG}	ℓ_{MAGMAR}	LR	$\widehat{\phi}$	$\widehat{\theta}$
25	1988	38	-143.719	-143.724	-0.008	0.625	-0.688
50	1991	35	-132.709	-131.288	2.840	0.574	-0.703
100	1996	30	-114.745	-116.919	-4.348	0.349	-0.890

Table 11: Influential-year sensitivity under the dynamic beta-binomial margin.

Masked year(s)	ℓ_{MAG}	ℓ_{MAGMAR}	LR	$\widehat{\phi}$	$\widehat{\theta}$
1975	-170.099	-168.964	2.271	0.700	-0.480
1974	-171.175	-168.975	4.399	0.678	-0.484
1975, 1974	-167.465	-166.795	1.339	0.684	-0.502

E Direction-specific exploratory fits

Table 12: Direction-specific interval fits under static beta-binomial margins.

Direction	Model	$\log L$	AIC	BIC	$\widehat{\phi}$	$\widehat{\theta}$
Downgrade	MAG	-152.604	311.208	317.637	—	0.941
Downgrade	MAGMAR	-154.218	316.436	325.009	0.199	0.148
Upgrade	MAG	-159.072	324.145	330.574	—	0.884
Upgrade	MAGMAR	-157.428	322.855	331.428	0.554	-0.229

Declarations of Interest

The author reports no conflicts of interest and is solely responsible for the content and writing of the paper.

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